Joint Quantile Regression by Optimal Transport*

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Abstract

Often in econological inference literature and program evaluation with cross-sectional models, empirical researchers face data that do not have complete information of individuals whose missing parts are marginally known but not jointly observed. In this paper, we propose a novel estimator constructed from a procedure analogous to EM algorithm for quantile regression models whose marginal distributions are known but joint distribution is missing under integrated minimum distance framework using optimal transport methods for partial identification. Implementation is straightforward and does not require semiparametric estimation of nuisance kernels for resulting minimization problems as in past literature on two-sample statistics, reducing finite sample bias and computational burden and can be extended to multivariate cases easily. We propose a bootstrap method that can be used to construct estimates of confidence intervals consistently and show its asymptotic validity. Extension to the cases where other auxilary variables are available and dependent variables are vector-valued as in Carlier et al. (2016) is also discussed. Monte-Carlo experiments are presented in appendix to demonstrate finite-sample efficiency of our procedure.

*This paper is to be accompanied with Ahn (2017), which can be also found in author's website.

†The full version of the paper, with Monte-Carlo experiment results can be found in the author's website (https://chiyahn.github.io/) with the corresponding R package for implementation, `jointqreg`. All views expressed in this manuscript are solely those of the author and should not be attributed to the Bank of Canada.

1 Introduction

In a number of empirical studies in social science, researchers face additional aggregate data that have to be taken into account to control individual observations. This problem is somewhat classic in political science (Robinson, 1950), which is known as ecologoical inference. For instance, explaining voting behaviours requires controlling demographic information of individuals. A main challenge is that these data are often not jointly observable; one can estimate marginal distributions of demographic information of individuals and voting behaviours separately, but not jointly, due to confidentiality of the votes.

Previous literature have either relied on independence assumption for $Y \perp X$ or availability of instrument variables that are jointly observed with both X and Y. Commonly used estimators in this context are two-sample instrumental variable (TS-IV) estimators that are first proposed by Angrist and Krueger (1992) and its variants, two-sample two-stage least squares (TS-2SLS) estimators, whose properties have been studied extensively in Inoue and Solon (2010). However, they can been applied in linear regression framework only and, perhaps more importantly, still all require joint dataset (Y, Z) and (X, Z) which is not common in practice. For similar reason, analysis on models that require identification of both $F(Y|Z)$ and $F(X|Z)$ (Cross and Manski, 2002 and Molinari and Peski, 2006) have limited applicability.

The goal of this paper is to provide a robust estimation procedure to construct consistent estimates for confidence intervals in quantile regression models where only marginal distributions are observed in minimum distance framework using optimal transportation. We show validity of bootstrap procedure and provide a computationally savoury procedure that can be used to efficiently estimate confidence intervals in practice. Our procedure requires neither semiparametric estimator for the density for the resulting dual Monge problem as in Gretton et al. (2012).

The idea is as follows. Consider linear quantile models with $Y_i = \mathbf{X}_i\boldsymbol{\beta}(U_i)$ such that $\mathbf{X}_i \perp U_i$ with U_i following uniform distribution where Y_i and $\mathbf{X}_i$ are marginally observed. First, note that, by definition, Y_i should have the same distribution of $\mathbf{X}_i\boldsymbol{\beta}(U_i)$. Also, by independence condition $\mathbf{X}_i \perp U_i$, pairs $(\mathbf{X}_i, U_j)$ constructed from marginal distribution of $\mathbf{X}_i$ and U_j have the same distribution as the joint distribution of $(\mathbf{X}_i, U_i)$ so that $\mathbf{X}_i\boldsymbol{\beta}(U_i)$ can

be constructed by permutations of marginally observed $\mathbf{X}_i$ and U_j. Hence, the parameters of interests $\boldsymbol{\beta}$ can be estimated by $\boldsymbol{\beta}$ such that the distribution of $\mathbf{X}_i\boldsymbol{\beta}(U_j)$ is close to the distribution of Y_i. To measure the distance between two distributions, we use Wasserstein metrics.

The idea of estimating parameters based on distance between the empirical measure of the dependent variables and the one implied by the model is not entirely new. It can be dated back to a seminal work by Manski (1983) that proves consistency of parameters under independence condition. Subsequent work by Brown and Wegkamp (2002) draw asymptotic distribution, and its analysis has been extended by Torgovitsky (2017) in nonseparable IV settings to allow covariates to be endogeneous. However, as Gao and Galvao (2014) pointed out, inference procedure in the framework of Brown and Wegkamp (2002) heavily relies on finding optimal joint measure for covariates and dependent variables, whose construction is yet unknown. Gao and Galvao (2014) proceeds to transform the original integrated minimum distance problem as the dual of , and proceeds.

We contribute in literature in the following aspects: first, we propose a new estimation procedure that can be applied for statistical models where a marginal is known but its joint realization is unknown in quantile regression models. In contrast to previous literature that heavily relied on modeling based on dependencies on cohorts or existence of joint observations with auxilary variables (Angrist and Krueger, 1992, Cross and Manski, 2002, and Molinari and Peski, 2006), our new methodology does not impose such constraints. Especially, when there are auxilory variables that are independent with the missing variables, then the model is identified. Second, we provide a new approach on quantile regression models based on optimal transport, allowing multivariate quantile regression models without imposing any triangular structure assumption. Third, we extend previous literature on minimum distance estimators in optimal transport context to construct estimators for parameters that can be partially identified without computation of nuisance semiparametric estimators (Gao and Galvao, 2014) or user-defined construction of optimal distance measure (Brown and Wegkamp, 2002), reducing finite sample and selection bias. Fourth, we build a bridge between minumum distance methods and quantile regression models to show that minimum distance methods can be also applied in nonlinear econometric models.

The paper is organized as follows. In section 2, we present quantile regression models when jointly observed data are not available and discuss idnetification issues. We present a joint quantile regression (JQR) estimator in section 3 and describe its implementation under parametric settings and asymptotic properties. We extend the model in section 4 for the case when other auxilary variables are available. Monte-Carlo experiments are presented in section 5. The paper concludes in section 6, discussing future research directions.

The following notations are used throughout the paper: we use subscript 0 on an arbitrary parameter to denote it as population parameter. We use boldfaced letters for multivariate random variables. For a measure space $(\mathcal{X}, \mathcal{B}(\mathcal{X}), \mu)$ on $\mathcal{X} \subset \mathbb{R}^n$ with $n < \infty$ and the Borel σ-algebra $\mathcal{B}(\mathcal{X})$ on $\mathcal{X}$ with some measure μ, we denote $\operatorname{int} \mathcal{X}$ for the interior of $\mathcal{X}$ with respect to the topology induced by the Euclidean metric. $\mathcal{U}$ denotes the closed interval $[0, 1]$. We use the notation $\sim$ to mean "follows the distribution of." $U[0, 1]$ denotes the uniform distribution. We use $F_{\mathbf{X}}^{-1}(U)$ to denote the quantile function of a random variable $\mathbf{X}$, if well-defined. For some probability spaces $(\mathcal{X}, \mathcal{B}(\mathcal{X}), \mu)$ and $(\mathcal{Y}, \mathcal{B}(\mathcal{Y}), \nu)$, $\Gamma(\mu, \nu)$ denotes the set of couplings of μ and ν, i.e., the set of probability distributions over $\mathcal{X} \times \mathcal{Y}$ whose first and second marginal distributions are given by measures μ and ν respectively. $\mathbb{N}_N$ denotes the set of natural numbers up to N, i.e., $\mathbb{N}_N := \{i \in \mathbb{N}; i \leq N\}$. $\|\cdot\|$ is used to denote Euclidean norms.

2 Quantile regression without jointly observed data

In this section, we provide model specification and the population objective function that can be used to partially identify parameters that result in the quantile function having the same distribution as the one for dependent variables. We present some regularity conditions and illustrate identification issues and possible empirical applications by examples.

2.1 Model specification

Let $\{\mathbf{X}_i, Y_i\}_{i=1}^N$ be the N collection of $(1+p)$-multivariate random variables of covariates of interests and dependent variables on common support $\mathcal{X} \times \mathcal{Y} \subset \mathbb{R}^{(p+1)}$ with the corresponding Borel $\sigma-$algebra $B(\mathcal{X} \times \mathcal{Y})$ that are not necessarily jointly observed by an econometrician.

Throughout the paper, we consider the following quantile specification of Y_i conditional on $\mathbf{X}_i$:

$$Q_{Y|\mathbf{X}}(\mathbf{X}_i, U_i; \boldsymbol{\beta}) = \mathbf{G}(\mathbf{X}_i)'\boldsymbol{\beta}(U_i) \tag{1}$$

where $\mathbf{G} : \mathbb{R}^p \to \mathbb{R}^K$ is a vector-valued measurable function whose specification is known. Note that, when we set $\mathbf{G}(\mathbf{X})$ as the identity mapping, i.e., $\mathbf{G}(\mathbf{X}) := \mathbf{X}$, (1) becomes a canonical linear quantile linear function. Alternative choices of $\mathbf{G}$ include orthogonal polynomials or splines that can be used to approximate the quantile function uniformly with large samples and K in a series representation. For reference, see Chen (2007) for possible choices of the functionals in sieve literature and their properties.

We consider a model from the parametric family of models $\boldsymbol{\beta} \in \boldsymbol{\Theta}$ to make partial identification tractable; we discuss its parametric implementation with more details later in subsection 3.2. We denote $\nu_{\boldsymbol{\theta}}$ the measure induced by $G_{\boldsymbol{\theta}}(\mathbf{X}_i, U_j)$. It is not difficult to see that $\nu_{\boldsymbol{\theta}}$ is indeed a probability measure.

We now introduce the population criterion function defined by Wasserstein distance and briefly discuss how it can be used to determine the distance between the distribution of Y and $Q_{Y|\mathbf{X}}(\mathbf{X}, U)$. Throughout the paper, we use the following population criterion function:

$$M(\boldsymbol{\theta}) = \min_{\gamma \in \Gamma(\mu, \nu_{\boldsymbol{\theta}})} \iint |Y - Q_{Y|\mathbf{X}}(\mathbf{X}, U; \boldsymbol{\beta})| d\gamma \tag{2}$$

where $\Gamma(\mu, \nu_{\boldsymbol{\theta}})$ is the set of couplings (optimal transport plans) between μ, probability measure implied by Y, and $\nu_{\boldsymbol{\theta}}$, probability measure implied by $Q_{Y|\mathbf{X}}(\mathbf{X}, U; \boldsymbol{\beta})$. The population objective function $M(\boldsymbol{\theta})$ compares the distance between these two measures by finding optimal assignment between μ and $\nu_{\boldsymbol{\theta}}$ that yields matchings that have the least absolute deviance on average. In fact, $M(\boldsymbol{\theta})$ defines the norm between μ and $\nu_{\boldsymbol{\theta}}$, and is called the Kantorovich-Rubinstein distance in optimal transport literature (Villani, 2009, Chapter 6), i.e., $M(\boldsymbol{\theta}) = 0$ holds if and only if μ and $\nu_{\boldsymbol{\theta}}$ are identitcal almost surely. This might not be achievable since there are numerical errors coming from the fact that we are approximating $\boldsymbol{\beta}$ based on interpolation in finite points under parametric specification, which is discussed with more details in subsection 3.2. We define the corresponding identified set as follows:

Definition 2.1 (Identified set)**.** We define the identified set $\boldsymbol{\Theta}_I$ as

$$\boldsymbol{\Theta}_I := \{\boldsymbol{\theta}_I \in \boldsymbol{\Theta}; M(\boldsymbol{\theta}_I) \leq M(\boldsymbol{\theta}), \quad \forall \boldsymbol{\theta} \in \boldsymbol{\Theta}\} \tag{3}$$

which is the collection of $\theta \in \mathbf{\Theta}$ that yields the conditional quantile function that has the closest distribution with the one for dependent variables.

2.2 Examples

In this subsection, we discuss a few empirical examples where dependent variables and covariates are not jointly observed.

Examples 2.1 (Estimating cross-sectional quantile treatment effects with persistent heterogeneity)**.** Consider the case when an econometrician is interested in post-treatment effect with persistent heterogeneity for each agent into consideration, as in Autor (2003) and Imbens and Wooldridge (2009), so that lagged variables are taken into account. Let Y_{it} denote the variable of interest observed in period t from ith individual such that

$$Y_{i2} = \rho Y_{i1} + \mathbf{Z}_i \boldsymbol{\beta} + U_i$$

where $\mathbf{Z}_i$ is a vector of variables that describe the characteristics of ith individual and U_i iid follows some distribution. In this case, ρ captures the persistent heterogeneity of the individual i.

Suppose that the dataset is available in a cross-sectional form such that tracking the past observation of each individual is impossible. One remedy is to use group observations in cohorts by regions and use the cohort-mean as a proxy variable for the previous observation, as suggested in Deaton (1985) and then in Verbeek and Vella (2005), which is still used in practice (Dang et al., 2014). Nevertheless, this methodology can suffer from huge bias due to possible tail behaviours of observations within-cohort and cohort criteria selection problem across cohorts (Mackinnon and Webb, 2017). Also, clearly, using mean for the past observation does not fully capture heterogeneity across individuals in the past.

Examples 2.2 (Earning dynamics with firm heterogeneity)**.** Consider an earning regression that not only depends on an ith worker's characteristics $\mathbf{Z}_i$, but also on firm heterogeneity $\mathbf{X}_{im}$ in industry m such that the earning of ith worker Y_i in mth industry is determined by

$$Y_i = \mathbf{X}_{im} \boldsymbol{\beta}(U_i) + \mathbf{Z}_i \boldsymbol{\gamma}(U_i)$$

Suppose we do not have matched employer employee data, but observe $(Y_i, \mathbf{Z}_i)$ jointly and $\mathbf{X}_{j,m}$ marginally for each industry m. In this case, for each i, one can use the information about $\mathbf{Z}_i$, but not $\mathbf{X}_{im}$ as the firm ith worker belongs to is unknown.

2.3 Regularity condition and results

Now we present and discuss the following regularity assumptions used throughout the paper:

Assumption 2.1 (Compactness and finite-dimensionality of $\boldsymbol{\Theta}$)**.** $\boldsymbol{\Theta}$ is compact and finite-dimensional.

Assumption 2.2 (Compactness of $\mathcal{X} \times \mathcal{Y}$)**.** $\mathcal{X} \times \mathcal{Y}$ is a compact subset of $\mathbb{R}^{p+1}$.

Assumption 2.3 (Continuity of the quantile function)**.** For each $\boldsymbol{\beta} \in \boldsymbol{\Theta}$, $Q_{Y|\mathbf{X}}(\mathbf{X}, U; \boldsymbol{\beta})$ is continuous on $\mathcal{X} \times \mathcal{U}$ in Euclidean distance.

Assumption 2.4 (Quantile restriction with regularity conditions)**.** $\boldsymbol{\beta}(\cdot)$ is continuous on $[0, 1]$ and $Q_{Y|\mathbf{X}}(\mathbf{X}, \tau)$ is strictly increasing on τ, conditional on $\mathbf{X}$.

Assumption 2.1 is standard. Assumption 2.2 is needed to guarantee that the criterion function (4) is well-defined. Note that Assumption 2.3 does not allow discontinuity in distribution of $\mathcal{Y}$. If the observations for dependent variables have noticable different distribution across certain cohorts as studied in Deaton (1985), one can allow having multiple objective functions instead; details are discussed in subsection 3.4. Assumption 2.4 is a standard assumption employed in quantile regression literature.

In the following lemma, we show that the population criterion function (2) is well-defined and the identified set (3) is not empty.

Lemma 2.1. The criterion function $M(\boldsymbol{\beta})$ in (2) has the following properties:

1. For every $\boldsymbol{\beta} \in \boldsymbol{\Theta}$, the criterion function $M(\boldsymbol{\beta})$ is well-defined.

2. $M(\boldsymbol{\beta})$ is continuous in $\boldsymbol{\beta}$, and is differentiable almost everywhere.

3. $\boldsymbol{\Theta}_I$ is a non-empty set.

Proof. For brevity, for $\xi := (\mathbf{X}, U) \in \mathcal{X} \times \mathcal{U}$, we let $Q(\xi; \boldsymbol{\beta}) := Q_{Y|\mathbf{X}}(\mathbf{X}, U)$ denote the conditional quantile function of Y throughout the proof.

For the first part of the lemma, note that for each $\boldsymbol{\beta}$, the space generated by $Q(\xi; \boldsymbol{\beta})$ is compact, which directly follows by Assumption 2.2 and Assumption 2.3. Hence, for each $\boldsymbol{\beta} \in \boldsymbol{\Theta}$, $\mathcal{Y}$ and $\mathcal{Y}_{\boldsymbol{\beta}}$ are compact, where $\mathcal{Y}_{\boldsymbol{\beta}} := \{Q(\xi; \boldsymbol{\beta}); \xi \in \mathcal{X} \times \mathcal{U}\}$. The rest of the argument is standard; see Theorem 1.4 of Santambrogio (2015) for example.

To prove the second part of the lemma, note that by Assumption 2.2, Assumption 2.1 and Assumption 2.3, $Q(\xi; \boldsymbol{\beta})$ is $K-$Lipschitz in $\boldsymbol{\beta}$ with some constant $K > 0$. Thus, for all $\boldsymbol{\beta}, \boldsymbol{\beta}' \in \boldsymbol{\Theta}$, we have

$$|W(\mu, \nu_{\boldsymbol{\beta}}) - W(\mu, \nu_{\boldsymbol{\beta}'})| \leq W(\nu_{\boldsymbol{\beta}}, \nu_{\boldsymbol{\beta}'}) \leq \iint |Q(\xi; \boldsymbol{\beta}) - Q(\xi; \boldsymbol{\beta}')| \, d\gamma \leq K \|\boldsymbol{\beta} - \boldsymbol{\beta}'\|$$

for any $\gamma \in \Gamma(\nu_{\boldsymbol{\beta}}, \nu_{\boldsymbol{\beta}'})$, the first inequality follows from the fact that Wasserstein distance is a metric and triangle-inequality, and the second inequality follows by the definition of Wasserstein distances, and the third inequality follows from the claim above that $Q(\xi; \boldsymbol{\beta})$ is $K-$Lipschitz. This shows that $W(\mu, \nu_{\boldsymbol{\beta}})$, i.e., $M(\boldsymbol{\theta})$, is Lipschitz in $\boldsymbol{\theta}$ almost everywhere. By the Rademacher's theorem, this Lipschitz condtion also implies that $M(\boldsymbol{\theta})$ is almost everywhere differentiable as well.

The third part of the lemma follows from the second part of the lemma with Assumption 2.1. □

For the rest of the paper, we shall consider partial identification of the models since point identfication can generally fail. The following examples illustrate this in separable model settings:

Remarks 2.1. Consider the following separable model

$$Y_i = \beta_1 X_{i1} + \beta_2 X_{i2}$$

where X_{i1} and X_{i2} follow standard normal distribution and X_1 and X_2 are independent. We note that this specific model has been also studied Sgouropoulos et al. (2015). Consider the case where $\beta_1 = \beta_2 = \sqrt{2}$. In this case, we have that $Y_i \sim N(0, 4)$.

Even if there are more separable terms whose distributions might not be normal, point-identification is most likely going to fail. We can consider this separable model with coefficients as mixture models whose mixing distribution is not normalized to one, where each

separable term yields a component distribution for a finite normal mixture model. For instance, the simple example corresponds to the model with 2-component finite normal mixture in both location and scale whose mixing distribution is not normalized and the order of each component is fixed (i.e., X_{i1} and X_{i2} cannot be interchanged, as we observe realization from both components.). As suggested in classic literature on identification of parametric finite mixture models, Teicher (1963) and Lindsay and Roeder (1993), identifiability condition is most likely not going to hold unless we apply additional restrictions.

3 Joint quantile regression estimator

In this section, we provide the definition of joint quantile regression estimators and a procedure based on EM algorithm that can be used to efficiently estimate the model in (1). We study its asymptotic properties and propose a bootstrap method that can be used to construct confidence intervals based on the argument of Chernozhukov et al. (2007). Extension to the case where dependent variables are multivariate is also discussed.

3.1 Definition

We begin this subsection by providing the definition of the joint quantile regression estimators:

Definition 3.1. Let $\{Y_i\}_{i=1}^N$ and $\{\mathbf{X}_i\}_{i=1}^N$ be the obseravations that are not necessarily jointly observed. The joint quantile regression estimator under specification given in (1) is defined as $\widehat{\boldsymbol{\beta}} := \arg\inf_{\boldsymbol{\beta}\in\boldsymbol{\Theta}} M_N(\boldsymbol{\beta})$ where $M_N(\boldsymbol{\beta})$ is a finite sample version of the population criterion function in Equation 2:

$$M_N(\boldsymbol{\beta}) = \min_{\gamma\in\Gamma(\mu_N,\nu_{MN\boldsymbol{\beta}})} \iint |Y - Q_{Y|\mathbf{X}}(\mathbf{X}, U; \boldsymbol{\beta})| d\gamma \tag{4}$$

where μ_N denotes the empirical probability measure constructed by $\{Y_i\}_{i=1}^N$ and $\nu_{MN\boldsymbol{\beta}}$ denotes the empirical probability measure implied by $\{Q_{Y|\mathbf{X}}(\mathbf{X}_i, U_j; \boldsymbol{\beta})\}_{(i,j)\in\mathbb{N}_N\times\mathbb{N}_M}$.

To see that the criterion function (4) is well-defined and the JQR estimator, $\boldsymbol{\beta}$ exsits, note that the argument given for regularity conditions of the population objective function

in Lemma 2.1 also holds for the sample criterion function defined in (4) by exchanging the population marginal distributions in the set of copulings to empirical distributions.

In the remaining part of the subsection, we illustrate how JQR estimators provide matching between dependent variables and estimated realization of conditional quantile functions.

Examples 3.1 (Simple matching procedure)**.** To see how JQR estimators provide matching between dependent variables and covariates, consider the following simple linear model without error with uniform regressor across all quantile levels:

$$Y_i = X_i\beta \quad \text{where } \beta \in \mathbb{R}^+ \tag{5}$$

such that marginal data $\{Y_i\}_{i=1}^N, \{X_i\}_{i=1}^N \subset \mathbb{R}^+$ are available but not the complete joint dataset $\{Y_i, X_i\}_{i=1}^N$. Note that β is point-identified and can be consistently estimated by $\sum_{i=1}^N Y_i / \sum_{i=1}^N X_i$ from SLLN under some regularity conditions without reliance on JQR estimation. Also, in this case, the quantile function in (1) is uniform across all quantile levels conditional on the covariate X_i; hence, we can set $M = 1$ without loss of efficiency.

First, for every $\beta \in \mathbf{\Theta}$, by quantile representation of Wasserstein distance in one-dimensional case (Santambrogio, 2015, Proposition 2.17), (4) can be alterantively written as

$$M_N(\beta) = \int_0^1 |\widehat{F}^{-1}_{\mu_N}(\tau) - \widehat{F}^{-1}_{\nu_{N\beta}}(\tau)| d\tau \tag{6}$$

where $\widehat{F}^{-1}_{\mu_N}(\tau)$ and $\widehat{F}^{-1}_{\nu_N}(\tau)$ are empirical quantile functions constructed by $\{Y_i\}_{i=1}^N$ and $\{X_i\beta\}_{i=1}^N$. This can be also represented as

$$M_N(\beta) = \sum_{k=1}^N \left| Y_{N(l)} - X_{N(l)}\beta \right| \tag{7}$$

where $Y_{N(k)}$ and $X_{N(k)}\beta$ are lth order statistics of samples of $\{Y_i\}_{i=1}^N$ and $\{X_i\beta\}_{i=1}^N$ respectively.

The alternative representation in Example 3.1 provides intuitive description on how JQR estimators are computed by 2-step matching of observed dependent variables and empirical observations from the model implied by the parameter of interest β. Given some estimate β, first compute the optimal matching given by a mapping $\sigma : \mathbb{N}_N \to \mathbb{N}_N$ that yields

$(Y_i, X_{\sigma(i)}\beta)_{i=1}^N$ that minimizes the sample criterion function (4)[1]. Then, compute β that minimizes the sum of pairwise difference of the pairs found in the first step. In this example, estimation procedure reduces to 2-step as it suffices to find β that minimizes (7) after computing order statistics since the order of $X_i\beta$, which is going to be invariant for every $\beta > 0$.

Note that when covariates are multivariate the matching procedure above still remains to be valid. Even for the case when we have quantile functions that are not uniform across all quantile levels, i.e., when not imposing $M > 1$ results in misrepresentation of its distribution, computation of JQR estimation yields similar steps. In this case, the main difference is that single Y_i is matched with M observations of $\{Q_{Y|\mathbf{X}}(\mathbf{X}_j, U_k; \boldsymbol{\beta})\}_{(j,k)\in\mathbb{N}_N\times\mathbb{N}_M}$. This can be easily extended to the case when the numbers of observations of covariates X_i and dependent variables Y_i differ as well; details of the estimation procedure are provided in the next subsection.

3.2 Implementation

In this subsection, we consider parametric implementation of JQR estimators by linear interpolants to provide a tracktable framework for practical use. We then provide a recursive procedure that can be considered as an analogy of EM algorithm to efficiently compute JQR estimators and study its computational aspects analytically.

3.2.1 Parametric implementation

For practical use, building on Wei and Carroll (2009) and Arellano and Bonhomme (2016), we consider approximation of $\boldsymbol{\beta}(\cdot)$ by splines evenly divided by Q points, $\{\tau_q\}_{q=0}^{Q+1}$, i.e., $\tau_q := q/(Q+1)$. We consider linear splines for implementation throughout the paper for simplicity and its corresponding representation of the quantile function that can be used to efficiently solve the second step (10) in Algorithm 3.1[2]. Since $\boldsymbol{\beta}(\cdot)$ is continuous on a

[1]Note that this is exactly equivalent to finding pairs $(Y_i, X_{\sigma(i)})$ since we restrict our parameter of interests β to be non-negative and $X_j > 0$ for all $j \in \mathbb{N}_N$.

[2]In practical implementation in our R package `jointqreg`, we let the tail distributions for the cases where $U < \tau_1$ and $U > \tau_Q$ approximated by having the slope of $\boldsymbol{\beta}$ on $[0, \tau_2]$ and $[\tau_{Q-1}, 1]$ constant for simplicity. Note that consistency of the estimator under $Q \to \infty$ still remains valid.

compact domain by Assumption 2.4, linear spline approximation is sufficient to consistently estimate $\boldsymbol{\beta}$ uniformly as $Q \to \infty$. Some algebra show that $Q_{Y|\mathbf{X}}(\mathbf{X}, U; \boldsymbol{\beta}) = \mathbf{G}(\mathbf{X})\boldsymbol{\beta}(\tau)$ for $\tau \in [0, 1]$ can be approximated by

$$\begin{aligned} Q_{Y|\mathbf{X}}(\mathbf{X}, U; \boldsymbol{\beta}) &= \sum_{q=0}^{Q+1} \mathbf{G}(\mathbf{X})' \left[(Q+1)[\boldsymbol{\beta}(\tau_{q+1}) - \boldsymbol{\beta}(\tau_q)] \left(U - \frac{q}{Q+1} \right) + \boldsymbol{\beta}(\tau_q) \right] \\ &\qquad \times 1\left(\tau_q < U \leq \tau_{q+1}\right) \\ &= \sum_{q=0}^{Q+1} \mathbf{G}(\mathbf{X})' \left[(1 - C(U))\boldsymbol{\beta}(\tau_q) + C(U)\boldsymbol{\beta}(\tau_{q+1}) \right] 1\left[\tau_q < U \leq \tau_{q+1}\right] \end{aligned} \tag{8}$$

where $C(U)$ is defined as $C(U) := (Q+1)(U - q/(Q+1))$. The second representation in (8) provides a compact and efficient way to construct a design matrix that can be used to solve for $\boldsymbol{\beta}$ in (10) by expressing the quantile function in terms of the parameters of interests $\{\boldsymbol{\beta}(\tau_q)\}_{q=0}^{Q+1}$. Also, note that, under this spline representation, M, the number of observations used for U_i, does not have to be the same as the number of quantile levels for approximation.

3.2.2 Estimation procedure

Algorithm 3.1 (Computation of joint quantile regression estimator $\boldsymbol{\beta}$)**.** Let $\{U_m\}_{m=1}^{M}$ be samples independently drawn from uniform distribution. Given initial guess $\boldsymbol{\beta}^{(0)}$, repeat the following procedure until changes in (9) reach a predetermined criteria.

1. Find an optimal transportation plan, provided by a set-valued mapping $(\sigma_X(i), \sigma_U(i)) : \mathbb{N}_N \to \mathbb{N}_N \times \mathbb{N}_M$ whose values contain M elements such that

$$\sum_{i=1}^{N} \sum_{j=1}^{M} |Y_i - \mathbf{G}(\mathbf{X}_{(\sigma_X(i))_j})' \widehat{\boldsymbol{\beta}}^{(l)}(U_{(\sigma_U(i))_j})| \tag{9}$$

   is minimized.

2. Given the optimal transportation plan from the first step, compute M-estimator $\widehat{\boldsymbol{\beta}}^{(l+1)}$ such that

$$\widehat{\boldsymbol{\beta}}^{(l+1)} = \arg\min_{\boldsymbol{\beta} \in \Theta} \sum_{i=1}^{N} \sum_{j=1}^{M} |Y_i - \mathbf{G}(\mathbf{X}_{(\sigma_X(i))_j})\boldsymbol{\beta}(U_{(\sigma_U(i))_j})| \tag{10}$$

Implementation of matching procedure above is straightforward and is computationally efficient. This can be also thought as an analogy to EM algorithm if we consider

optimal transportation plan computed in (9) as latent variables. The first step can be computed by ordering and matching dependent variables and estimated quantile representations. After sorting $\{Y_i\}_{i=1}^N$ and $\{\mathbf{G}(\mathbf{X}_i)'\boldsymbol{\beta}^{(l)}(U_j)\}_{(i,j)\in\mathbb{N}_N\times\mathbb{N}_M}$, relabel them as $\{Y_{(i)}\}_{i=1}^N$ and $\{\mathbf{G}(\mathbf{X}_{(j)})'\boldsymbol{\beta}^{(l)}(U_{(j)})\}_{j=1}^{MN}$ by their orders respectively. By quantile representation as in (6), the Wasserstein distance is given by the sum of absolute pairwise difference between two observations in each matching. The main difference here is that the number of observations can be different (N and MN, respectively). It is not difficult to see that the resulting distance is equivalent as the matching procedure outlined above with having M duplicates for each dependent observation $\{Y_i\}_{i=1}^N$ since each ith dependent observation, $Y_{(i)}$, is going to have a weight of $1/N$ on quantile levels of $[(i-1)/N, i/N)$, whose range contains M observations from $\{\mathbf{G}(\mathbf{X}_{(j)})'\boldsymbol{\beta}^{(l)}(U_{(j)})\}_{j=1}^{MN}$. Hence, the set-valued mapping $(\sigma_X(i), \sigma_U(i))$ in the first step is explicitly given by $\{\mathbf{G}(\mathbf{X}_{(M(i-1)+h)})'\boldsymbol{\beta}^{(l)}(U_{(M(i-1)+h)})\}_{h=1}^M$.

Note that the objective function (10) in the second step of the procedure is convex under linear interpolant implementation given by (8), which guarantees the existence of solution. Also, the second equation of (8) provides representation of the quantile function that can transform the resulting minimization problem as a linear least absolute deviation (LAD) model whose parameters of interests are $\{\boldsymbol{\beta}(\tau_q)\}_{q=0}^{Q+1}$, which can be solved by using statistical packages widely available in public.

The following theorem shows that the objective function in (10) decreases in each iteration, showing validity of the procedure outlined in Algorithm 3.1.

Proposition 3.1. Algorithm 3.1 monotonically decreases the objective function in (4). That is, $M_N(\boldsymbol{\beta}^{(l+1)}) \leq M_N(\boldsymbol{\beta}^{(l)})$.

In practice, multiple local minima may exist. We recommend using different candidates for initial parameters $\{\boldsymbol{\beta}^{(0)}\}$ as suggested by other papers that consider computational aspects of EM algorithm; see Biernacki et al. (2003) for example.

The next theorem states our JQR estimators converge to the identified set in (3) as the number of sample size increases for partial identification.

Proposition 3.2. A JQR estimator, $\widehat{\boldsymbol{\beta}} := \arg\min_{\boldsymbol{\beta}\in\boldsymbol{\Theta}} M_N(\boldsymbol{\beta})$, converges to the identified set in (3) in half-Hausdorff distance as N increases.

Proof. The argument closely follows Theorem 3.1 in Chernozhukov et al. (2007). We denote $\boldsymbol{\Theta}_I^\epsilon$ to denote the $\epsilon-$expansion of $\boldsymbol{\Theta}_I$, i.e., $\boldsymbol{\Theta}_I^\epsilon := \{\boldsymbol{\beta} \in \boldsymbol{\Theta}; \sup_{\boldsymbol{\theta}_0 \in \boldsymbol{\Theta}_I} \|\boldsymbol{\beta} - \boldsymbol{\beta}_0\| \leq \epsilon\}$

Fix $\epsilon > 0$. Let $\boldsymbol{\beta}_0 \in \boldsymbol{\Theta}$ be a member in the identified set, which is not empty by Lemma 2.1, and let $\boldsymbol{\beta}_N$ be the JQR estimator given N samples. By continuity of the objective function from Lemma 2.1, there exists $\delta(\epsilon) > 0$ such that $\boldsymbol{\beta} \in \boldsymbol{\Theta} \setminus \boldsymbol{\Theta}_I^\epsilon$ implies that $M(\boldsymbol{\beta}) - M(\boldsymbol{\beta}_0) \geq \delta(\epsilon)$. Also, note that $\boldsymbol{\beta} \in \boldsymbol{\Theta} \setminus \boldsymbol{\Theta}_I^\epsilon$ holds if and only if $\inf_{\boldsymbol{\beta} \in \boldsymbol{\Theta}_I} \|\boldsymbol{\beta} - \boldsymbol{\beta}_0\| > \epsilon$.

Hence, we have that

$$\begin{aligned}
\mathbf{P}\left(\inf_{\boldsymbol{\beta} \in \boldsymbol{\Theta}_I} \|\boldsymbol{\beta}_N - \boldsymbol{\beta}\| > \epsilon\right) &= \mathbf{P}\left(M(\boldsymbol{\beta}_N) - M(\boldsymbol{\beta}_0) \geq \delta(\epsilon)\right) \\
&= \mathbf{P}\left(M(\boldsymbol{\beta}_N) - M_N(\boldsymbol{\beta}_N) + M_N(\boldsymbol{\beta}_N) - M(\boldsymbol{\beta}_0) \geq \delta(\epsilon)\right) \\
&\leq \mathbf{P}\left(M(\boldsymbol{\beta}_N) - M_N(\boldsymbol{\beta}_N) + M_N(\boldsymbol{\beta}_0) - M(\boldsymbol{\beta}_0) \geq \delta(\epsilon)\right) \\
&\leq \mathbf{P}\left(2 \sup_{\boldsymbol{\beta} \in \boldsymbol{\Theta}} |M(\boldsymbol{\beta}) - M_N(\boldsymbol{\beta})|\right) \xrightarrow[N \to \infty]{} 0
\end{aligned}$$

where the third line of ineqaulity follows by the definition of $\boldsymbol{\beta}_N$ and the last convergence result follows by compactness of $\boldsymbol{\Theta}$ and continuity of Wasserstein distances (See Theorem 6.9 in Villani, 2009, for example.). □

We discuss practical implmentation of confidence interval construction using bootstrap in subsection 3.3.

3.3 Constructing confidence regions

Suppose that an econometrician is interested in constructing a confidence interval for $\boldsymbol{\beta}$ of interest. We propose using bootstrap construction for confidence intervals drawn from partial identification literature, which has been first introduced and recommended in Chernozhukov et al. (2007) and Iliberto and Tamer (2009), implemented by both subsampling and parametric bootstrap methods. Implementation is straightforward and intuitive.

The estimation procedure for both methods is as follows; difference lies only in bootstrap sample generation in the second step, which is discussed in the next paragraph.

Algorithm 3.2 (Consturction of confidence interval of $\boldsymbol{\beta}$)**.** Let N data set $\{Y_i\}_{i=1}^N$ and $\{X_i\}_{i=1}^N$ be given, with the JQR estimator $\widehat{\boldsymbol{\beta}}$ from the samples.

1. Generate B number of bootstrap samples with N observations for each bth sample, denoted as $\{Y_i^{(b)}\}_{i=1}^N$ and $\{X_i^{(b)}\}_{i=1}^N$ with corresponding sample criterion function $M_N^{(b)}$ and minimizers $\widehat{\boldsymbol{\beta}}^{(b)}$.

2. Let p be the quantile level of confidence interval of interest. For each bth bootstrap sample generated in the second step, compute $\widehat{d}_{(b)} = M_N^{(b)}(\widehat{\boldsymbol{\beta}}) - M_N^{(b)}(\widehat{\boldsymbol{\beta}}^{(b)})$.

3. Choose pth percentile of $\widehat{d}_{(b)}$, and denote it as $\widehat{c}_p$. Then, the confidence region for $\boldsymbol{\beta}$ is given by $\{\boldsymbol{\beta} \in \boldsymbol{\Theta}; M_N(\boldsymbol{\beta}) - M_N(\widehat{\boldsymbol{\beta}}) \leq \widehat{c}_p\}$.

We suggest some bootstrap methods that can be used to generate samples in Algorithm 3.2. The first bootstrap method is using subsampling as suggested by Chernozhukov et al. (2007); in this case, bth bootstrap sample is generated by randomly drawn observations from the original data set. This method can be attractive in situations when the number of observations, N, is large, as in models with cross-sectional data. Another approach would be to use parametric bootstrap methods; use the JQR estimator $\widehat{\boldsymbol{\beta}}$ computed from the initial sample observations to construct the quantile function $\widehat{Q}_{Y|\mathbf{X}}(\mathbf{X}, U)$, and generate N samples per each pth bootstrap sample from generating dataset $\{\widehat{Q}_{Y|\mathbf{X}}(\mathbf{X}_i^{(p)}, U_i^{(p)})\}_{i=1}^N$ $\{\mathbf{X}_i^{(p)}\}_{i=1}^N$ for Y and $\mathbf{X}$ observations respectively by randomly sampling $\mathbf{X}_i^p$ and drawing U_i^p from uniform distribution independently.

3.4 Extension

3.4.1 JQR estimators for vector quantile regression

Interplay of savings and consumption can play a huge role in models of earning process. Usually, empirical analysis is carried by vector autoregressive process that is captued by multivariate versions of ordinary least square methods. We can extend the analysis to quantile regression models in multivariate settings too, as in Carlier et al. (2016). In this subsection, we briefly discuss computation of JQR estimators under multivariate settings and its practical implementation.

Let $\{\mathbf{Y}_i\}_{i=1}^N$ be $d-$dimensional N observations of dependent variables with $\{\mathbf{X}_i\})_{i=1}^N$ regressors that are not jointly observed with some $\{U_m\}_{m=1}^M$ drawn from uniform distribution.

We relabel $(i, m) \in \mathbb{R}_N \times \mathbb{N}_M$ to a single index $k \in \mathbb{R}_{MN}$. We consider the following quantile specification of Y_i conditional on $\mathbf{X}_i$:

$$\mathbf{Q}_{\mathbf{Y}|\mathbf{X}}(\mathbf{X}_i, U_i; \boldsymbol{\beta}) = \mathbf{H}(\mathbf{X}_i)'\boldsymbol{\beta}(U_i) \tag{11}$$

where $\mathbf{H} : \mathbb{R}^p \to \mathbb{R}^K \times \mathbb{R}^D$ is a matrix-valued measurable function whose specification is known. For construction, proof of existence of multivariate conditional quantile functions, and regularity conditions, refer to Carlier et al. (2016) in details. Especially, Carlier et al. (2016) show that the distribution of $(\mathbf{Q}_{\mathbf{Y}|\mathbf{X}}(\mathbf{X}, U; \boldsymbol{\beta}), \mathbf{X})$ is the same as that of $(\mathbf{Y}, \mathbf{X})$ when the model is correctly specified. Also, note that we do not have to impose a triangular structure as in Arellano and Bonhomme (2016), thus allowing more diversity in selection of $\mathbf{H}(\mathbf{X})$.

The population criterion function (2) is going to be almost identical as the notion of Wasserstein distance is kept in multivariate cases too:

$$M(\boldsymbol{\theta}) = \min_{\gamma \in \Gamma(\mu, \nu_{\boldsymbol{\theta}})} \iint \|\mathbf{Y} - \mathbf{Q}_{Y|\mathbf{X}}(\mathbf{X}, U; \boldsymbol{\beta})\| d\gamma \tag{12}$$

with its corresponding sample objective function:

$$M_N(\boldsymbol{\beta}) = \min_{\gamma \in \Gamma(\mu_N, \nu_{MN\boldsymbol{\beta}})} \iint \|\mathbf{Y} - \mathbf{Q}_{Y|\mathbf{X}}(\mathbf{X}, U; \boldsymbol{\beta})\| d\gamma \tag{13}$$

where $\mu, \nu_{\boldsymbol{\theta}}$ are probability measures induced by $\mathbf{Y}$ and $\mathbf{Q}_{Y|\mathbf{X}}(\mathbf{X}, U; \boldsymbol{\beta})$ respectively, and $\mu_N, \nu_{MN\boldsymbol{\beta}}$ are empirical probability measures induced by their samples.

The only difference with univariate cases introduced in section 3 is the norm in the integrand since we consider the case where dependent variables are vector-valued. Note that this multivariate structure does not allow quantile representation as in univariate cases in (6). In fact, this makes a huge contrast in estimation procedure: the matching procedure is not going to hold in this case with finite samples. This makes the first step of iterative procedure proposed in Algorithm 3.1 replaced by solving the corresponding earth mover's distance (EMD) problem to solve for (13) given $\boldsymbol{\beta}$. Details are as follows:

Algorithm 3.3 (Computation of joint quantile regression estimator $\boldsymbol{\beta}$, multivariate case)**.** Given initial guess $\boldsymbol{\beta}^{(0)}$, repeat the following procedure until changes in (14) reach a predetermined criteria.

1. Solve the earth mover's problem to find optimal $\{\gamma_{ik}^{(l)}\}_{(i,k)\in\mathbb{N}_N\times\mathbb{N}_{MN}}$ such that

$$\sum_{i=1}^{N}\sum_{k=1}^{MN}\gamma_{ik}^{(l)}|\mathbf{Y}_i-\mathbf{H}(\mathbf{X}_k)'\widehat{\boldsymbol{\beta}}^{(l)}(U_k)| \tag{14}$$

   is minimized, where $\sum_{k=1}^{MN}\gamma_{ik}=1$ for all $i\in\mathbb{N}_N$ and $\sum_{i=1}^{N}\gamma_{ik}=1$ for all $k\in\mathbb{N}_{MN}$.

2. Given the optimal transportation plan from the first step, compute M-estimator $\widehat{\boldsymbol{\beta}}^{(l+1)}$ such that

$$\widehat{\boldsymbol{\beta}}^{(l+1)}=\arg\min_{\boldsymbol{\beta}\in\Theta}\sum_{i=1}^{N}\sum_{k=1}^{MN}\gamma_{ik}^{(l)}|\mathbf{Y}_i-\mathbf{H}(\mathbf{X}_k)'\boldsymbol{\beta}(U_k)| \tag{15}$$

Solving for pairs $\{\gamma_{ik}^{(l)}\}_{(i,k)\in\mathbb{N}_N\times\mathbb{N}_{MN}}$ in the first step of algorithm (14) can take considerable amount of time due to the resulting linear programming problem, unlike in one-dimensional problem. For details, see Galichon (2016). The second step in (15) is analogous to the one-dimensional case, and it can be solved by a number of statistical packages available in public.

It is also worth mentioning that validity of the above repeated 2-step procedure and asymptotic properties of JQR estimators still hold; as mentioned in the proof provided in Proposition 3.1 and Proposition 3.2, the identical argument for multivariate cases can be drawn since our proof does not require variables to be univariate, and Wasserstein distance can be defined in vector-valued random variables as well.

3.4.2 Estimation under multiple objective functions

Suppose that one wants to estimate quantile regression models where observations differ significantly in terms of the distribution of their corresponding covariates. This is setup can be especially useful in estimation of parametric nonlinear difference-in-difference models. For instance, consider the case when an econometrican wants to estimate quantile regression models for both treatment group and control group to account for the influence of a certain binary treatment. Assuming that treatment assignment is random, to compute the quantile treatment effects $\boldsymbol{\delta}(U)$, she makes the following specification for the conditional quantile function:

$$Q_{Y|\mathbf{X},D}(U,\mathbf{X},D)=\mathbf{X}\boldsymbol{\beta}(U)+D\boldsymbol{\delta}(U)$$

where D is a binary variable such that $D = 1$ if an agent is in treatment group and $D = 0$ if not, and $\mathbf{X}$ is the vector of covariates that characterize each participant's demographic information. This model fits into our framework, but due to the presence of D, this yields two different objective functions for the treated and control groups. Also, the distribution of $\mathbf{X}$ can be different in the two groups accounting for unobservable correlation with D.

More generally, consider the case when each observation is grouped by some common characteristics as g up to G groups. WLOG, assume that all groups have the same number of observations N and observations are all generated by the same conditional quantile function $Q_{Y|\mathbf{X}}(\mathbf{X}, U) = \mathbf{G}(\mathbf{X})\boldsymbol{\beta}(U)$. Relabeling observations by adding a superscript for the group each observation belongs to, we have $\{Y_i^g\}_{i=1}^N$ and $\{\mathbf{X}_i^g\}_{i=1}^N$ sample. Then, each gth sample yields gth objective function $M_N^g(\boldsymbol{\beta})$.

The resulting estimation procedure remains the same. In fact, Algorithm 3.1 is valid by replacing $M_N(\boldsymbol{\beta})$ by $\sum_{g=1}^G M_N^g(\boldsymbol{\beta})$. Note that matching procedure in the first step (9) has to be solved for each subproblem $M_N^g(\boldsymbol{\beta})$. On the other hand, solving for the second step (10) yields a LAD problem with GMN observations as they all share the same parameters of interests $\boldsymbol{\beta}$.

In numerical experiments section 5, we find that this discretization process provides sharper confidence intervals when the distribution of covariates (thus dependent variables) differ by each group, even after fixing the number of observations. We discuss potential explanation of this phenomena in section 6, which is an interesting future research topic by itself.

4 Joint quantile regression with other covariates

In this section, we extend the model to the case where we have extra covariates $\mathbf{Z}_i$ that are jointly observed with Y_i in tuples $(Y_i, \mathbf{Z}_i)_{i=1}^N$. As in previous sections, we do not assume that each tuple is jointly observed with $\mathbf{X}_i$. This makes our analysis different from previous literature that requires either joint dataset for both (Y, Z) and (X, Z) (Angrist and Krueger, 1992) or distributional information about $F(Y|Z)$ and $F(X|Z)$ (Cross and Manski, 2002 and Molinari and Peski, 2006).

4.1 Model

Let $\{\mathbf{X}_i, \mathbf{Z}_i, Y_i\}_{i=1}^T$ be the collection of $(p+q+1)$-multivariate random variables of regressors, auxilary variables, and dependent variables of interests on common support $\mathcal{X} \times \mathcal{Z} \times \mathcal{Y} \subset \mathbb{R}^{(p+q+1)}$ with the corresponding Borel $\sigma-$algebra $B(\mathcal{X} \times \mathcal{Z} \times \mathcal{Y})$. As in section 2, we consider the following conditional quantile specification of Y:

$$Q_{Y|\mathbf{X},\mathbf{Z}}(\mathbf{X}_i, \mathbf{Z}_i, U_i; \boldsymbol{\beta}) = \mathbf{G}(\mathbf{X}_i, \mathbf{Z}_i)'\boldsymbol{\beta}(U_i) \tag{16}$$

where $\mathbf{G} : \mathbb{R}^p \times \mathbb{R}^q \to \mathbb{R}^K$ is a vector-valued measurable function whose specification is known and $\boldsymbol{\beta}$ allows parametric implementation as in subsection 3.2.

4.2 Special case: when X and Z are independent

It is worth mentioning that the model is identified if $\mathbf{X}$ and $\mathbf{Z}$ are independent. This setup can be found in a number of empirical studies where treatment are distributed randomly by a policymaker, especially in empirical program evaluation literature using lottery programs like Imbens et al. (2001), Jacob and Ludwig (2012), and Cesarini et al. (2016). Note that we do not require any panel data settings here, hence providing more applicability.

First, note that the conditional density $f(\mathbf{X}|Y, \mathbf{Z}; \boldsymbol{\beta})$ corresponds to $f(Y|\mathbf{X}, \mathbf{Z}; \boldsymbol{\beta})f(\mathbf{X})f(\mathbf{Z})/f(Y, \mathbf{Z})$. In finite-sample settings, $f(\mathbf{X})$, $f(\mathbf{Z})$, and $f(Y, \mathbf{Z})$ can be consistently estimated from the data. $f(Y|\mathbf{X}, \mathbf{Z}; \boldsymbol{\beta})$ is also identified by the specification given in (16):

$$f(Y|\mathbf{X}, \mathbf{Z}) = \lim_{\delta \to 0} \frac{\delta}{\mathbf{G}(\mathbf{X}, \mathbf{Z})' \left[\boldsymbol{\beta}(\tau^* + \delta) - \boldsymbol{\beta}(\tau^*)\right]}$$

where $\tau^* := \{\tau \in (0,1); Y = \mathbf{G}(\mathbf{X}, \mathbf{Z})'\boldsymbol{\beta}(\tau)\}$ since the conditional density of Y is the inverse of the derivative of its conditional quantile function. Note that this can be also consistently estimated by computing the inverse of the numerical derivative of the conditional quantile function.

Hence, one can use $f(\mathbf{X}|Y, \mathbf{Z}; \boldsymbol{\beta})$ to draw $(Y_i, \mathbf{Z}_i, \mathbf{X}_j)_{(i,j) \in \mathbb{N}_N \times \mathbb{N}_M}$ jointly. Estimation procedure is then rather straightfoward. From stochastic EM algorithm by Nielsen (2000), we can 1. sample $(Y_i, \mathbf{Z}_i, \mathbf{X}_j)_{(i,j) \in \mathbb{N}_N \times \mathbb{N}_M}$ by the estimated $f(\mathbf{X}|Y, \mathbf{Z}; \boldsymbol{\beta})$ from Metropolis-Hastings and 2. update $\boldsymbol{\beta}$ by solving the resulting quantile estimating equation using the sample $(Y_i, \mathbf{Z}_i, \mathbf{X}_j)_{(i,j) \in \mathbb{N}_N \times \mathbb{N}_M}$ drawn from the first step, and repeat the two steps above until

convergence. Note that the resulting objective function is sum of quantile check functions, which is not log-likelihood function as originally studied in Nielsen (2000). In this case, the argument used in (Arellano and Bonhomme, 2016, Section 3.3) for extension to quantile estimating equations holds. The only difference is that $\mathbf{X}$, which corresponds to individual heterogeneity η in errors-in-variables framework studied in Arellano and Bonhomme (2016), is directly observed from data; hence, the same argument holds by replacing $f_{\eta|X}(\eta|X)$ by $f(\mathbf{X})f(\mathbf{Z})$.

4.3 Estimation procedure

Unfortunately, in most cases, we do not have additional information about dependency between $\mathbf{X}$ and $\mathbf{Z}$. In program evaluation literature, this relation is captured by the generalized propensity score function in continuous treatment settings (Hirano and Imbens, 2004), $f_{vX|\mathbf{Z}}(\mathbf{X})$, the density of treatment X conditional on covariates Z. When complete data $(\mathbf{X}_i, \mathbf{Z}_i)_{i=1}^N$ are observed as complete pairs, $f_{\mathbf{X}|\mathbf{Z}}(\mathbf{X})$ can be consistently constructed. This is exactly the situation covered by a number of works in past literature; see Angrist and Krueger (1992), Cross and Manski (2002), and Molinari and Peski (2006) for examples. On the other hand, note that this exactly fits into our framework as studied in section 2; we have marginal observations of $\mathbf{X}$ and $\mathbf{Z}$, but missing joint observations of $(\mathbf{X}, \mathbf{Z})$.

For simplicity, we assume that the auxilary random variable X is one-dimensional in this subsection. We discuss implementation in multi-dimensional cases at the end of this chapter. Assuming that X is univariate, we specify the quantile propensity score function as in section 2 in the following form:

$$Q_{X|\mathbf{Z}}(\mathbf{Z}, V_1; \boldsymbol{\delta}) = \mathbf{H}(\mathbf{Z})\boldsymbol{\delta}(V_1) \tag{17}$$

where $\mathbf{Z} \perp V_1$, $V_1 \sim U[0,1]$ with the following conditional quantile regression function of Y given $(X, \mathbf{Z})$:

$$Q_{Y|X,\mathbf{Z}}(X, \mathbf{Z}, V_2; \boldsymbol{\beta}) = \mathbf{G}(\mathbf{X}, Z)\boldsymbol{\beta}(V_2) \tag{18}$$

where $(X, \mathbf{Z}) \perp V_2$, $V_2 \sim U[0,1]$. which results in the 2-step estimator that consists of the follwing procedure:

First, we estimate the quantile generalized propensity score function $Q_{X|\mathbf{Z}}(\mathbf{Z}, U; \boldsymbol{\delta})$. This is done by findjing JQR for the conditional quantile function $Q_{X|\mathbf{Z}}(\mathbf{Z}, V_1; \boldsymbol{\delta}) = \mathbf{H}(\mathbf{Z})\boldsymbol{\delta}(V_1)$, given marginally observed $\{X_i\}_{i=1}^N$ and $\{\mathbf{Z}_i\}_{i=1}^N$. This yields the first-step JQR estimator $\widehat{\boldsymbol{\delta}}(\cdot)$ and the following imputed dataset: for each ith tuple, construct M triplets $(Y_i, \widehat{X}_{ij}, \mathbf{Z}_i)_{j=1}^M$ based on imputed samples where $\widehat{X}_{ij} = \mathbf{H}(\mathbf{Z}_i)\widehat{\boldsymbol{\delta}}(j/(M+1))$, which yields MN joint observations. Second, using the new dataset $(Y_i, \widehat{\mathbf{X}}_{ij}, Z_i)_{i,j}^{MN}$, perform ordinary quantile regression (Koenker, 2005) on $Q_{Y|\mathbf{X},\mathbf{Z}}(\mathbf{X}, \mathbf{Z}, V_2) = \mathbf{G}(\mathbf{X}, \mathbf{Z})\boldsymbol{\beta}(V_2)$. This yields the parameter of interests, $\boldsymbol{\beta}(\cdot)$.

If $\mathbf{X}$ is a $d-$dimensional multivariate random variable, once can compute the JQR estimator for vector quantile regressions as in subsection 3.4 to find $\boldsymbol{\delta}(\cdot)$ in (17) while allowing $\mathbf{H}(\mathbf{Z})$ be matrix valued. The remaining procedure is identical.

5 Numerical experiments[3]

6 Conclusion

When joint observations of dependent and covariates are not available, the best way to track dependency relation between them is to find the parameters that will let the estimated model have the distribution closest to the distribution of the dependent variables. We use quantile regression to model dependence between covariates and dependent variables and have built tracktable estimation procedure that can be used to consistently (partially) identify the parameters of interests. Despite the fact that there is no other extra information about dependency, as expected by recent improvement in theoretical bounds on partial identification for ecological inference (Fan et al., 2014), our simulation results show that JQR estimators provide sharp confidence intervals. We also discuss estimation procedure when there are auxilary variables by constructing quantile generalized propensity score functions.

There are a number of possible applications we have not discussed here in details yet. We can extend the analysis while having an extra time-axis to capture persistent within-cohort heterogeneity in an extension of Callaway et al. (2017) in repeated cross-sectional

[3]This part has been removed due to length limit for writing sample submission. The complete version of this paper is available in our website: https://chiyahn.github.io/.

settings allowing dynamic persistent heterogeneity. Our quantile specification can be thought as copula stability assumption from Callaway et al. (2017), as the dependency relation is going to be determined and stable along time period by the specified quantile function, but identification results from Callaway et al. (2017) suggest that panel-data settings are necessary to draw conditions. What if we have only cross-sectional data, so that the present observations are not jointly observed with their past history? Even in this worse case we can apply our partial identification tools to compute confidence intervals consistently, which can still yield meaningful results.

This dependency modeling also suggests a number of theoretical aspects that can be used to build bridges between finite mixture models and copula theory. Numerical experiments show that estimation based on grouping observations by discretization based on different distributions of $\mathbf{X}$ leads to a sharper confidence interval, as compared to the case where all observations are treated as one group. Interestingly, this discretization process usually provides better-than-expected Fréchet-Hoeffding bounds from implied copulas based on conditional expectations. It can be thought of as a simple analogy to rank conditions used in identification literature in mixture models (Allman et al., 2009 and Bonhomme et al., 2016). We expect that it can be studied under a similar framework, thereby suggesting possible links between finite mixture models and dependency bounds in copula theory.

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